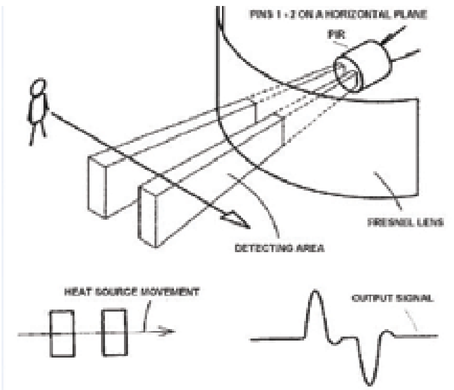
## Detect intruders

When you're out of your house, an intruder may try to gain entry into it. You don't need to install a camera to constantly monitor for break-ins. Leave the task to sensors called 'PIR sensors', which we'll be using in a while to make an intruder detector. Sometimes called 'PID' (Passive Infrared Detector), the PIR sensor works by detecting infrared rays. You might have heard of or seen an infrared camera that shows the body of a human in red, blue and green colour. The camera is able to capture this because the human is emitting infrared rays, which is, in fact, heat that a human body radiates.

Let's see how a PIR sensor works. Refer to the image showing the working of a PIR sensor.




Working of a PIR sensor

Every PIR sensor has two slots in it, each containing an infrared radiation detector. A special lens called 'Fresnel' lens (that's used in lighthouses near coasts) is placed in front of the two sensors to increase its area of reach. In the absence of a human, both sensors detect the same amount of infrared radiation. When a human passes in front of the sensor, the first sensor detects heat and the second doesn't detect any heat, leading to a difference in output of the two sensors. When the human moves to the other side, the second sensor detects heat, but the first doesn't detect any heat then. Thus, from the change in heat patterns, a PIR sensor detects the presence of a human. To know how PIR sensors work in detail, visit this link: *http://dgit.in/1H2Q64b* .